

DESCRIPTIVE STATISTICS

Mean	$\bar{x} = \frac{1}{n} \sum x_i$	<code>mean(x)</code>
Median	middle value (data ordered)	<code>median(x)</code>
Variance	$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$	<code>var(x)</code>
Std dev	$s = \sqrt{s^2}$	<code>sd(x)</code>
Range	max - min	<code>range(x)</code>
IQR	$Q_3 - Q_1$ (the fourth spread)	<code>IQR(x, type = 2)</code>
Outlier rule: below $Q_1 - 1.5 \text{ IQR}$ or above $Q_3 + 1.5 \text{ IQR}$.		

PROBABILITY RULES

Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Complement	$P(A^c) = 1 - P(A)$
Mutually exclusive	$P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B)$
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Multiplication	$P(A \cap B) = P(A B) P(B)$
Independence	$P(A \cap B) = P(A) P(B)$; also $P(A B) = P(A)$
Bayes' Rule	$P(A B) = \frac{P(B A) P(A)}{P(B)}$
Total Probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Equally likely outcomes: $P(A) = N(A)/N$.	

COUNTING

Product rule	k stages: $n_1 n_2 \cdots n_k$
Combinations	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ <code>choose(n,k)</code>
Permutations	$P(n, k) = \frac{n!}{(n-k)!}$ <code>factorial(n)</code>

DISCRETE RANDOM VARIABLES

Valid pmf: $p(x) \geq 0$, $\sum_x p(x) = 1$.	cdf: $F(x) = P(X \leq x) = \sum_{y \leq x} p(y)$.
Expected value	$E(X) = \mu = \sum_x x p(x)$
Variance	$V(X) = \sigma^2 = \sum_x (x - \mu)^2 p(x)$
Shortcut	$V(X) = E(X^2) - [E(X)]^2$
Std dev	$\sigma = \sqrt{V(X)}$
Linear rules	$E(aX + b) = aE(X) + b$, $V(aX + b) = a^2 V(X)$, $\sigma_{aX+b} = a \sigma_X$

CONTINUOUS RANDOM VARIABLES

Valid pdf: $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$.	$P(X = c) = 0$, so $\leq$ and $<$ are interchangeable.
cdf	$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$
Probability	$P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$
pdf from cdf	$f(x) = F'(x)$
$E(X)$	$\int_{-\infty}^{\infty} x f(x) dx$
$E[h(X)]$	$\int_{-\infty}^{\infty} h(x) f(x) dx$
$V(X)$	$E(X^2) - [E(X)]^2$
Percentile η_p	Solve $F(\eta_p) = p$; median solves $F(m) = 0.5$

NAMED DISTRIBUTIONS

Distribution	pmf or pdf	$E(X)$, $V(X)$	R Code
Binomial $X \sim \text{Bin}(n, p)$	$p(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, \dots, n$	np $np(1-p)$	<code>dbinom(x,n,p)</code> <code>pbinom(x,n,p)</code> <code>qbinom(p,n,p)</code>
Poisson $X \sim \text{Pois}(\mu)$	$p(x) = \frac{e^{-\mu} \mu^x}{x!}$, $x = 0, 1, 2, \dots$ Over an interval of length t : $\mu = \lambda t$	μ μ	<code>dpois(x,mu)</code> <code>ppois(x,mu)</code> <code>qpois(p,mu)</code>
Normal $X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$ Standardize $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$, unstandardize $x = \mu + z\sigma$	μ σ^2	<code>pnorm(x,mu,sigma)</code> <code>qnorm(p,mu,sigma)</code>
Exponential $T \sim \text{Exp}(\lambda)$	$f(t) = \lambda e^{-\lambda t}$, $F(t) = 1 - e^{-\lambda t}$, $t \geq 0$ $P(T > t) = e^{-\lambda t}$; memoryless, $P(T > s + t T > s) = P(T > t)$	$1/\lambda$ $1/\lambda^2$	<code>pexp(x,rate)</code> <code>qexp(p,rate)</code>

In the R Code column: **d** = pmf or pdf, **p** = $P(X \leq x)$, **q** = quantile.

SAMPLING DISTRIBUTIONS & THE CLT

Sample mean	$E(\bar{X}) = \mu$, $SE(\bar{X}) = \sigma/\sqrt{n}$ (estimated by $s/\sqrt{n}$)
Normal population	$\bar{X} \sim N(\mu, \sigma^2/n)$ exactly, for any n
CLT	For $n > 40$, $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$
Standardized	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
Sample proportion	$\hat{p} = X/n$, $E(\hat{p}) = p$, $SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$
Conditions	Means: $n > 40$. Proportions: $np \geq 10$ and $n(1-p) \geq 10$

HYPOTHESIS TEST STEPS

- Hypotheses:** state H_0 and H_a in terms of population parameters.
- Conditions:** verify the assumptions for the chosen test.
- Test statistic:** compute t , z , or F from the data.
- P-value:** use the R code in the Inference Procedures table for the row you chose and the form of H_a . For an F test, `1-pf(F,df1,df2)`.
- Decision:** reject H_0 if the P-value is at or below α , otherwise fail to reject.
- Conclusion:** interpret in context.

REGRESSION ASSUMPTIONS (LINE)

Linear relationship
Independent observations
Normal residuals
Equal variance

ONE-WAY ANOVA

$H_0: \mu_1 = \mu_2 = \dots = \mu_I$ versus H_a : at least two differ.

Source	Sum of squares	df	Mean square
Treatments	$SSTr = \sum_i J_i (\bar{y}_i - \bar{y}_{..})^2$	$I - 1$	$MSTr = \frac{SSTr}{I-1}$
Error	$SSE = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2$	$n - I$	$MSE = \frac{SSE}{n-I}$
Total	$SST = SSTr + SSE$	$n - 1$	

$F = \frac{MSTr}{MSE} \sim F_{I-1, n-I}$, always an upper-tail test
`1 - pf(F, I-1, n-I)`

I = number of groups, n = total observations, J = observations per group

Conditions

- Independent random samples
- Approximately normal populations
- Equal variances: largest s no more than about twice the smallest

INFERENCE PROCEDURES

Parameter	Conditions	Test statistic	Confidence interval	Critical value	P-value by H_a
μ σ known, or σ unknown and $n > 40$	Random sample σ known and population normal, or $n > 40$	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ use s for σ when $n > 40$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	<code>qnorm(1-a/2)</code>	$> 1 - \text{pnorm}(z)$ $< \text{pnorm}(z)$ $\neq 2 * \text{pnorm}(-\text{abs}(z))$
μ σ unknown, $n \leq 40$	Random sample Population approx. normal	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$	$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$	<code>qt(1-a/2, n-1)</code>	$> 1 - \text{pt}(t, n-1)$ $< \text{pt}(t, n-1)$ $\neq 2 * \text{pt}(-\text{abs}(t), n-1)$
p	Random sample $np_0 \geq 10$ $n(1 - p_0) \geq 10$	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$ the test uses p_0	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$ the interval uses $\hat{p}$	<code>qnorm(1-a/2)</code>	$> 1 - \text{pnorm}(z)$ $< \text{pnorm}(z)$ $\neq 2 * \text{pnorm}(-\text{abs}(z))$
$\mu_1 - \mu_2$ independent	Two independent samples Both normal, or both $n > 40$	$t = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \sim t_\nu$	$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, \nu} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	<code>qt(1-a/2, nu)</code>	$> 1 - \text{pt}(t, \text{nu})$ $< \text{pt}(t, \text{nu})$ $\neq 2 * \text{pt}(-\text{abs}(t), \text{nu})$
μ_D paired	Random sample of matched pairs, $n_1 = n_2$ Differences approx. normal when $n \leq 40$	$t = \frac{\bar{d} - \Delta_0}{s_D/\sqrt{n}} \sim t_{n-1}$ on the differences $d_i = x_i - y_i$	$\bar{d} \pm t_{\alpha/2, n-1} \frac{s_D}{\sqrt{n}}$	<code>qt(1-a/2, n-1)</code>	$> 1 - \text{pt}(t, n-1)$ $< \text{pt}(t, n-1)$ $\neq 2 * \text{pt}(-\text{abs}(t), n-1)$
$p_1 - p_2$	Two independent samples $n_i \hat{p}_i \geq 10$ $n_i(1 - \hat{p}_i) \geq 10$	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ pooled $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$	$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$ not pooled	<code>qnorm(1-a/2)</code>	$> 1 - \text{pnorm}(z)$ $< \text{pnorm}(z)$ $\neq 2 * \text{pnorm}(-\text{abs}(z))$

Two-sample df: $\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$, truncated down.

Sample size (CI for μ , σ known): $n = \left(\frac{2 z_{\alpha/2} \sigma}{w} \right)^2$, where w is the width of the interval. Always round up. Quadrupling n halves the width.